RISKNET: CONNECTING THE DOTS IN FRAUD INVESTIGATION

ABSTRACT

Anti-Money Laundering (AML) systems generate thousands of daily alerts, but nearly 85–95% of them are false positives. Because analysts typically investigate these alerts in isolation, they struggle to distinguish coordinated fraud networks from coincidental matches. This leads to delayed investigations, inconsistent decision-making, and high workloads.

RiskNet addresses this challenge as a specialized fraud investigation and case-management platform. Instead of acting as a raw detection engine, it helps analysts investigate existing alerts through relationship-driven analysis. RiskNet enriches every incoming alert by constructing a graph of shared devices, IP addresses, contact information, and identity attributes. An Isolation Forest model then processes graph features—such as degree centrality, community membership, and suspicious neighbourhood connectivity—to generate an explainable prioritization score. Rather than navigating a traditional, text-heavy list, analysts work directly within an interactive relationship graph that visualizes connected entities. The platform features an ML-prioritized queue, detailed case views, fluid graph traversal, and complete audit logging of analyst actions (such as clearing, escalating, or reporting) to ensure a transparent, compliant decision trail.

The backend is built with Java, Spring Boot, and RESTful APIs, utilizing Neo4j for native graph analytics and PostgreSQL to manage user accounts, case records, and audit logs. A Python-based pipeline runs the Isolation Forest model, while the React frontend leverages Cytoscape.js for interactive network exploration. The system is evaluated using a standardized, industry-representative bank fraud dataset augmented with synthetic relationship attributes like shared device and contact identifiers. Inspired by enterprise platforms like NICE Actimize and Feedzai, RiskNet differentiates itself by making the visual relationship graph, rather than a flat alert list, the primary investigative workspace.